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Beyond the Gaze: Using Eye-Tracking and Surveys to Understand VR Stories



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Abstract:
Virtual reality (VR) allows users to freely direct their attention within immersive environments, which creates challenges for narrative-driven experiences, as the users might miss important story elements. This study investigates whether eye-tracking data, as a measure of visual attention and registered gaze, reflects the same narrative engagement as post-experience surveys when analysing how players follow a story in VR. For this study, a narrative-driven prototype was developed in Unity using the Meta Quest Pro headset and an eye-tracking logging system. Ten participants experienced the prototype while their gaze behaviour was being recorded, followed by a post-experience survey focused on perceived object importance and narrative recall. The results showed differences between objective eye-tracking behaviours and subjective recall. The objects with high dwell times were not always perceived as important, while key narrative objects sometimes received limited recorded gaze time while being remembered by participants. These findings indicate that eye-tracking data and post-experience surveys may capture different aspects of narrative engagement in VR experiences. Given the limited sample size of ten participants, these findings are exploratory and should be interpreted with caution.

Keywords: Virtual Reality, Eye Tracking, Narrative Engagement, Immersive Storytelling

Abstrakt:
Virtuell verklighet (VR) tillåter användare att fritt rikta sin uppmärksamhet inom immersiva miljöer, vilket skapar utmaningar för berättelsedrivna upplevelser, eftersom användarna kan missa viktiga berättelseelement. Denna studie undersöker om ögonspårningsdata, som ett mått på visuell uppmärksamhet och registrerad blick, återspeglar samma narrativa engagemang som i enkäter efter upplevelsen, speciellt i hur spelaren följer en berättelse i VR. För denna studie utvecklades en narrativt driven prototyp i Unity med hjälp av headsetet Meta Quest Pro och ett ögonspårningsloggsystem. Tio deltagare upplevde prototypen medan deras blickbeteende registrerades, följt av ett formulär efter upplevelsen, vilket fokuserade på narrativ återkallelse och upplevd viktighet av objekt. Resultaten visade skillnader mellan objektiva ögonspårningsbeteenden och subjektiv återkallelse. Objekt med höga uppehållstider uppfattades inte alltid som viktiga, medan viktiga narrativa objekt ibland fick begränsad registrerad blicktid och deltagarna kom ihåg dessa objekt. Dessa fynd indikerar att ögonspårningsdata och formulär efter upplevelsen kan fånga olika aspekter av narrativt engagemang i VR-upplevelser.

Nyckelord: Virtuell verklighet, Ögonspårning, Narrativt engagemang, Immersivt berättande

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1. Introduction:

Virtual reality (VR) has become an increasingly important tool for interactive storytelling. Unlike traditional media, VR enables immersive experiences in which users actively engage with narrative elements. VR also allows users to direct their attention within a three-dimensional space, like in the real world, but this introduces new challenges in understanding how narratives are followed and perceived. For this reason, it is essential for developers, designers and researchers to understand how players engage and interpret story elements in these environments (Adhanom et al., 2023).

For the purpose of this study, narrative engagement is operationalised as the degree to which participants visually attend to and subsequently recall narrative elements within the VR environment. This is measured through three indicators: accumulated gaze dwell time on predefined objects, participant ratings of perceived object importance, and open-ended narrative recall. These indicators are not assumed to capture narrative engagement in its entirety, but rather to provide complementary perspectives on how participants interact with and interpret story elements during and after the experience.

Traditionally, post-experience surveys have been used to evaluate the narrative understanding and user experience. These methods rely on the participants to reflect on their own experience after it has occurred. This provides insight into perceived engagement, emotional response and interpretation. However, post-experience surveys may be limited by subjective interpretation (Neuert & Lenzner, 2016).

Eye tracking provides an alternative route by capturing real-time data about where users direct their visual attention during an experience. This makes it possible to study how users engage with narrative elements as the story unfolds, providing objective measurements of gaze behaviour and attention distribution. For VR, this becomes highly relevant as the users are free to look around and may miss important narrative cues depending on how their attention is guided (Moreno-Arjonilla et al., 2024).

Despite this, it remains unclear to what extent eye-tracking data aligns with self-reported experiences. Post-experience surveys capture what users remember and interpret, while eye-tracking measures what users look at (Duchowski, 2017). When comparing the data, these methods may represent different aspects of the experience, as behavioural and self-reported measures do not always align in user experience research (Kivikangas et al., 2011), raising questions about how well they correspond in analyses of narrative engagement.

This study was designed to investigate the relationship between eye-tracking data and post-experience surveys in the context of narrative-driven VR. Specifically, it examines how these methods align when analysing how players follow a story. By comparing subjective reports with objective gaze data, this research aims to contribute to a better understanding of how narrative engagement can be measured in immersive environments. Based on this, the study addresses the following research question: *Does eye tracking, as a measure of visual attention and registered gaze, capture the same aspects of narrative engagement as post-experience surveys when studying how players follow a story in VR?*

The following article is structured as follows. First, previous research related to attention guidance, narrative engagement and eye-tracking in VR is presented. After this, the methodology and implementation of the VR experience are described. The results from both the post-experience surveys

and eye tracking data are then presented and discussed in relation to the research question. Finally, the article concludes with limitations, reflections and suggestions for future work.

2. Background and Related Work:

2.1 Narrative Engagement in Virtual Reality

In regular movies and games, the creator controls the camera. They decide exactly what it is that the audience will see. If something important happens in a movie, the camera points right at it (Mateer, 2017) However, VR is different.

The player is placed inside a 360-degree environment where they control their own view by moving their head. This freedom changes how stories are told. It becomes challenging to ensure players actually notice the story elements. A developer can build a highly detailed scene, but the player might direct their attention toward irrelevant areas of the environment (Adhanom et al., 2023). This challenge is particularly evident in cinematic VR, where players frequently miss key narrative cues depending on where their attention happens to be directed (Nielsen et al., 2016).

A significant aspect of why VR is effective is something called presence. Slater and Sanchez-Vives (2016) split this into two parts. The first part is something called “Place illusion”. This is the feeling of actually standing in the virtual room. The user knows they are wearing a headset, but their brain tells them they are in a physical space. The second part is “Plausibility illusion”, which means that the events happening around them feel real and believable. If a virtual phone rings, the user reacts to it like a real phone. When both of these illusions work together, the brain treats the virtual world almost like reality.

This freedom however creates a major problem for storytelling. Players can look anywhere at any time, so they might easily miss important things (Sheikh et al., 2016). If a door opens behind a player, but they are looking elsewhere, they will miss that visual information completely. This means that following a story in VR depends on where the player happens to look at the exact time an event triggers.

Because of that, the developers have to adapt. They cannot rely on players sitting still and watching. They have to figure out how to get the player’s attention, but they also have to do it without taking away the player’s control. (Jerald, 2015).

If a game takes over the camera and forces the player to look at the door, the player can get motion sickness. (Rebenitsch & Owen, 2016). It also completely ruins the feeling of actually being there (presence) which, as earlier mentioned, is a core reason as to why VR works as well as it does. This means that designers have to use other natural ways to guide the player’s eyes towards the right objects.

2.2 Guiding User Attention in Immersive Environments

Since developers cannot force the camera to move, they have to use different cues instead. Cues are natural hints that tell the player’s brain where to look. Auditory cues represent one of the most

effective modalities for directing user attention. In real life, when an individual perceives an auditory cue from behind, they instinctively orient themselves toward the source. VR uses spatial audio to do the exact same thing (Pausch et al., 1996).

If a character starts talking from the left side of a room, the player will turn their head toward the voice. This mechanism exhibits high efficacy because human hearing covers 360 degrees (Serafin et al., 2018) Human vision is restricted to the immediate forward field of view, whereas the auditory system encompasses a 360-degree range, allowing sound to guide visual orientation.

Lighting is another natural guide. The human eye is automatically drawn to light (Sundstedt et al., 2008). If a virtual room is mostly dark and one specific area lights up, the player will look at that specific light. Designers use this to highlight important things. For example, they might put a spotlight on a door or make an item on a table glow. This shows the player where to look without actually telling them. (Cater et al., 2003).

Movement also catches our attention fast. A bird flying across the sky, or something falling off a shelf, will make the player look in that direction. The brain tracks movement to figure out what is happening in the environment. (Itti & Koch, 2001).

Furthermore, developers can use a physical feeling which is called haptic feedback. This means physical vibrations (Burdea, 1996). It can, for example, be a controller buzzing or a wearable suit vibrating. If a player feels a sudden rumble on their back, they will immediately look around to see what caused it. (Jerald, 2015).

By combining these tools, designers can control the flow of the story. They can play a sound while an object flashes, and lead the player step by step. However, even with good audio and visual cues, players still miss things. They might hear a sound but look at the wrong object, or they might ignore the hint completely. And that is why developers actually need to measure where players are looking, instead of hoping that the cues worked.

2.3 Eye Tracking as an Objective Measure of Visual Attention

To know if the visual cues actually work, guessing is not enough. Objective data is required. This requirement highlights the utility and relevance of eye-tracking. Many modern VR headsets have eye tracking integrated directly into the hardware (Koulieris et al., 2019). They use small cameras to watch the user's pupils. This tells the system exactly where the person is looking in the 3D space. It provides researchers with empirical, quantitative data for analysis. (Duchowski, 2017).

Before eye tracking became standard, developers often relied on head tracking. They drew an invisible line from the centre of the headset to see what the player was facing. This is not however an accurate measurement of attention. People do not always point their heads directly at what they are looking at (Sitzmann et al., 2018) Eye tracking solves this by measuring the actual pupil movement.

But to understand this data, we have to look at how human eyes actually move. Our eyes do not move smoothly across a room. Instead, they make very fast jumps. These jumps are called saccades. Between the jumps, the eyes stop and focus for a fraction of a second. This stop is called a fixation. The human brain only really processes clear visual information during a fixation. During a jump, we

are actually largely unable to process visual input. So when researchers study attention, they are mostly looking for these fixations (Holmqvist et al., 2011).

To catch these movements, the hardware needs to be fast. This speed is called the sampling rate. It is measured in Hertz (Hz). A normal research eye tracker might run at 60 or 120 Hz. That means it logs the eye position 60 or 120 times every single second. (Duchowski, 2017).

But if that sample rate is too low, it creates a big problem. A player might quickly glance at an object and look away before the system even takes its next snapshot. The data will show that they never looked at it at all.

There is another piece to this puzzle. The software needs to know what the user is looking at, not just the exact coordinates. In 3D game engines like Unity, developers put invisible 3D boxes around objects. These are usually called colliders (Clay et al., 2019). When the user's gaze hits this invisible box, the system logs a hit. In eye tracking research, these target areas are called "Areas of Interest" or "AOIs". (Holmqvist et al., 2011).

The problem with AOIs however, is that if the invisible box is the exact same size as a small object, it requires extreme precision to get hit. The player has to look dead centre at the item. If they look slightly to the side, the system misses it (Holmqvist et al., 2011), or if they spot the object in their peripheral vision while grabbing it. It will not register anything. Because of this, researchers often make AOIs much larger than the physical objects (Orquin et al., 2016). If the target area is too small or the logging speed is too slow, the data will miss quick interactions completely. Another problem with this, however, is that when you increase the size of AOIs, it might suddenly start logging the object even when they are not looking at it.

2.4 The Gap Between Objective Data and Subjective Recall

Post-experience surveys are the most common way to test if a VR story works. Researchers ask the players what they saw and how they felt after the experience was over. However, this method has a notable limitation, it relies entirely on human memory.

Human memory is not a video recording. When a player finishes a VR experience, their brain has already filtered the information. This is called memory bias (Schacter, 1999). The brain only keeps information that feels important or has an emotional impact. In psychology, this is called saliency (Itti & Koch, 2001). If a loud alarm goes off, it is highly salient. The player will definitely remember it. But if they spend a lot of time looking at a normal wall or a floor, such non-salient information is typically discarded from cognitive memory. It does not matter to the story. (Bylinskii et al., 2015).

This creates a serious gap. What a player actually looks at is often very different from what they say they looked at.

Eye tracking fixes part of this problem. It provides quantitative metrics that reveal actual physiological gaze patterns of where the eyes moved. But eye tracking has its own limits. It cannot tell us if the player actually understood what they saw. A player might stare right at an important clue, but their mind might be thinking about something else. The eye tracker logs a hit, but the player still misses the story. By putting the objective data and the subjective surveys right next to each other, developers can finally see the actual gap between looking and remembering.

3. Methodology:

3.1 Research Methods

This study employed a mixed-methods approach in combining qualitative post-experience survey data with quantitative eye-tracking data. The eye-tracking data was collected through a custom-built eye-tracking logging system. This study design aimed to examine how objective measures of visual attention relate to subjective interpretations of narrative understanding, to evaluate how these methods align when analysing user engagement in virtual reality.

The research was conducted using a narrative-driven VR prototype developed in Unity specifically for this study. Participants experienced a sequence of interactive story events, which were designed to guide attention towards key narrative elements while also still allowing free exploration of the environment. Eye-tracking data was collected through a custom-built logging system, while subjective experiences were captured through post-experience surveys.

3.1.1 Research Design

The study was designed to measure the relationship between visual attention and narrative perception in VR. A narrative-driven VR prototype was created where participants were exposed to predefined narrative events involving objects such as a telephone, a light source, a painting and a teddy bear, each of which played a role in conveying the narrative.

Visual attention data was recorded using a custom-built Unity system that logged when participants directed their gaze toward any of the objects within the VR space. This allowed the system to track which narrative elements were observed at what time during the experience, and also for how long the participant looked at said object.

The design of the experience was based on the need to create situations where users might miss important narrative cues. For example, certain events were triggered based on the duration of the user looking at the object, which requires participants to visually focus on specific objects for a defined period of time. This allowed for the measurement of whether participants actually looked at the elements that were important for the story. This also enabled the collection of structured data regarding whether participants visually engaged with key narrative elements during the experience.

3.1.2 Participants

The study included 10 participants. The gender distribution was 7 males and 3 females. The group's age range was 19 to 39 years, with the majority of participants being 21.

- **Experience Level:** All participants but one had prior experience with VR, with self-reported skill levels ranging from beginner to advanced.
- **Selection:** Participants were recruited from a student body and were “fresh eyes” who had no prior knowledge of the project’s specific narrative elements to ensure genuine reactions to attention-guiding results.

Participant	Age	Gender	VR Experience
P1	21	Male	Advanced
P2	25	Male	Intermediate
P3	21	Male	Beginner
P4	21	Male	Advanced
P5	21	Male	Intermediate
P6	26	Male	Intermediate
P7	39	Male	Beginner
P8	21	Female	Intermediate
P9	21	Female	Advanced
P10	19	Female	Never used before

Table 1. Overview of Participant Demographics and Prior Experience.

Ethical Considerations: Before participation, all participants received the necessary information about the study's purpose and provided informed consent**.** Participation was voluntary, and all participants could withdraw at any time. The collected eye-tracking logs and survey responses were anonymised before analysis. No identifiable information was included in the final dataset or reported in the study.

3.1.3 Data Collection Methods

1. **Quantitative gaze logging**

Objective visual attention was captured through a custom-built Unity system and saved as individual logs for each participant. The following data points were recorded through the logging system:

- **Timestamp:** The exact time an object was observed.
- **Object ID:** The name of the specific collider hit by the gaze (e.g, “Phone”, “Teddy Bear”, “Painting”).

2. **Qualitative post-experience surveys**

Immediately after playing the experience, the participants completed a survey. This survey captured:

- **Importance:** Participants rated the importance of the narrative object on a scale of 1 (not important) to 5 (super important).
- **Recall Accuracy:** Open-ended questions were asked, such as having participants describe the story and list what caught their attention.
- **Guiding cues:** Participants identified what they felt guided them, such as sound and light

The custom eye-tracking system operated at a sampling rate of 2hz, meaning gaze position was recorded twice per second. This represents a notable technical constraint, as rapid eye movements such as saccades occur significantly faster than this interval and were therefore frequently not captured by the logging system.

Dwell time was calculated as an accumulated total rather than requiring continuous fixation. The system counted the number of gaze hits registered per object and multiplied this by the average interval between measurements (0.5 seconds), yielding a cumulative dwell time value. For objects where one or more participants registered zero gaze hits, those participants were excluded from the per-object average to avoid artificially deflating the mean.

Area of interest boundaries were defined by Unity colliders that closely matched the geometry of each object and were not intentionally enlarged.

Consequently, the system required relatively precise gaze alignment to register a hit, meaning peripheral gaze directed towards the edges of objects was frequently not captured.

The Meta Quest Pro includes built-in hardware-level eye tracking sensors which all of the participants used.

3.1.4 Data Analysis

The quantitative eye-tracking logs were analysed by calculating the average dwell times for each object across all participants. Then the values were visualised through graphs and scatter plots to identify patterns between the eye-tracking behaviour and perceived narrative importance.

The qualitative survey responses were analysed through thematic grouping, mostly focusing on recurring descriptions related to narrative recall, attention guidance and perceived object importance. After this, the survey responses were interpreted alongside the eye-tracking data to identify similarities and differences between the objective eye-tracking behaviour and the subjective participant experiences.

3.2 Design Methods

3.2.1 VR experience

For this study, a narrative-driven VR experience was developed as an experimental artefact. The experience took place in a living room environment and presented a sequence of events which were designed to guide the participant's attention.

Key narrative elements included a lamp button, a painting, a telephone and a teddy bear (*see figure 1*). Combined with these objects were audio cues, lighting changes, and interaction mechanisms that initiated the progression of events.

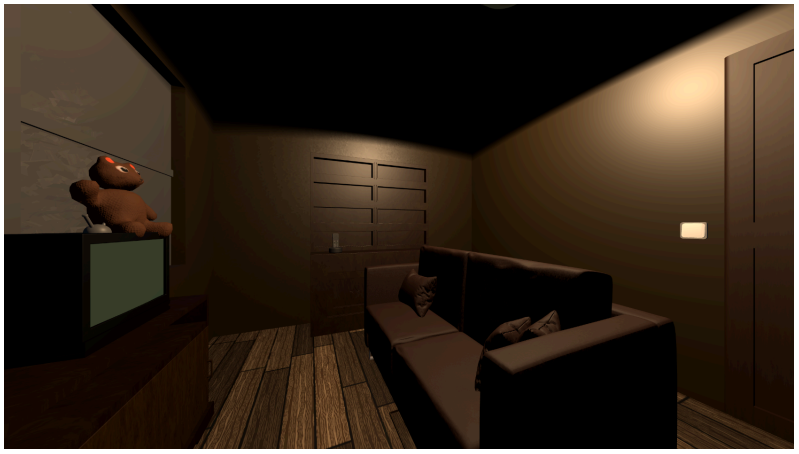


Figure 1. Overview of the narrative-driven environment showcasing the primary interactive objects.

The premise of the experience was designed to balance user freedom with guided attention, allowing participants to explore the space freely while still being directed toward important narrative moments through environmental cues and eye-tracking-based triggers.

The experience served as the primary medium for collecting eye-tracking data and the subjective user responses.

The narrative is set in a living room where a series of unsettling events unfold in a sequence. The participant begins with a lamp button in a dark room. This serves as the initial trigger and activates a telephone ringing sequence.

Following this, the painting plays whispering audio, requiring the participant to approach it in order to progress. After this, a shadow appears at the window which when the player looks at disappears. Following this, the lights go out and the teddy bear grows in size. When the participant directs their gaze toward it, its head is revealed to have rotated 180 degrees before returning to its original position.

The final narrative event is triggered when the television activates and displays the instruction to “Push the Sofa”, after which the participant pushes the sofa to reveal a hatch, completing the experience.

Throughout the sequence, the teddy bear functions as both an ambient atmospheric object and a secondary navigational aid, using a pointing gesture to direct the participant toward the next event. While it contributes to the overall narrative tone, it does not serve as a primary progression trigger.

3.2.2 Development Process

The VR experience was developed iteratively using an agile workflow inspired by the Scrum methodology. The development of the experience was structured into week-long sprints, where each iteration focused on implementing, testing and refining a specific part of the experience.

Internal testing was regularly done throughout the development process to ensure that the narrative triggers and eye-tracking-based events were behaving as intended. This process allowed for the continuous refinement of both the narrative flow and the technical implementation, which helped gauge how effectively the attention guiding mechanisms functioned in practice.

3.3 Project Methods

3.3.1 Equipment and Materials

- **Hardware:** This study used the Meta Quest Pro headset, which was chosen specifically for its integrated eye-tracking sensors
- **Software:** The environment was built in Unity using the Meta All-in-One SDK and the XR Interaction Toolkit.
- **Narrative Logic**: A "Story Manager" took care of the progression of the narrative. It utilised gaze triggers, meaning certain events, for example, the telephone teleporting, would only initiate once the system confirmed that the user had not looked at the telephone for at least 3 seconds.

3.3.2 Procedure

Each participant took part in the study individually. When starting, the participants get a brief introduction on how to interact with objects in VR and get told to freely walk around and experience the experience on their terms.

During the experience, the participant was able to walk around freely in the environment and explore the narrative elements as well as the sequence of eye-tracking interactions that would get triggered. Eye-tracking data was being recorded continuously while the participant was playing.

After completing the VR experience, the participant completed a post-experience survey. The survey focused on their recall of the perceived importance of objects, narrative elements and what they felt guided them through the experience. Each session lasted approximately 25 minutes, including both the VR experience and the survey. For an overview see figure 2:

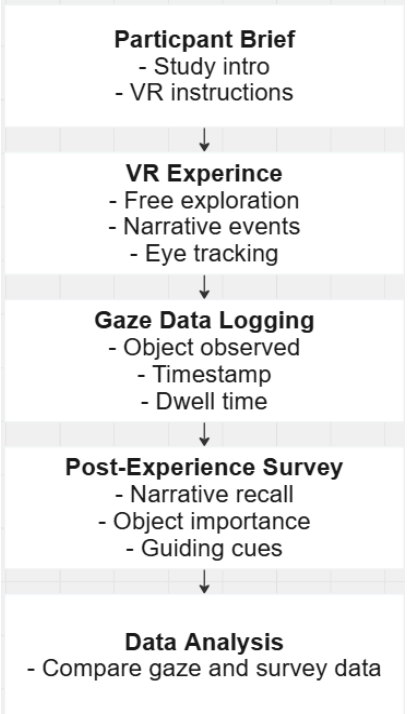


Figure 2. An overview of the study procedure.

3.3.3 Pilot Study

Before the main data collection, a pilot study was conducted to test the functionality of the VR experience and the reliability of the custom-built eye-tracking logging system. The purpose of the pilot study was to identify potential issues related to pacing, the flow of the interactions and the eye-tracking logging system, so it was not to collect final research data.

It was carried out with two participants recruited from a student environment. The participants had no prior knowledge of the project's narrative structure or intended outcomes, and they were asked to complete the VR experience under the same conditions as planned for the main study. During the pilot sessions, attention was mostly given to system stability. This process included verifying whether the eye-tracking-based interactions were being correctly registered and, consequently, if the narrative elements triggered as intended.. But the most attention was given to pacing, so if the narrative elements did not feel too fast or too slow.

After the participants had played the experience, the data was collected from the eye-tracking logger. Following the sessions, data was retrieved from the eye tracking logger to evaluate the overall functionality of the system. This included verifying that timestamps aligned with the in-experience event, that objects were correctly recorded and that the data was consistently recorded for later analysis.

Overall, the pilot study served as a validation step for both the technical implementation and the experimental design of the eye-tracking events and ensured that the system was suitable for structured data collection for the main study.

4. Results:

4.1 Objective Visual Attention (Eye-Tracking Data)

This section will present the data collected from the eye tracking logs. The tracking system recorded how long the participants looked at specific objects. This is called dwell time.

The data shows a substantial difference between the objects. The couch had the highest average dwell time. Participants looked at it for an average of 82.5 seconds. The ceiling was second, with an average of 26.0 seconds.

On the other hand, the objects that actually drove the story forward got very little visual attention. The phone only registered an average dwell time of 0.5 seconds. The teddy bear showed another interesting result. The system only registered 3 out of 10 participants looking directly at it. For those three, the average time was 15.2 seconds. But for the other seven participants, the hardware registered no gaze. This indicates that gaze registration was not consistent across all narrative objects, particularly for smaller elements.

4.2 Subjective Recall and Experience (Survey Data)

After the VR experience, the participants answered a survey. We asked them what they felt guided their attention through the story. Every single participant stated that sound guided them. 5 out of 10

participants (50%) said that light helped them find where to look. Movement cues and haptics were reported by fewer participants, n < 3 in both cases (see figure 3).

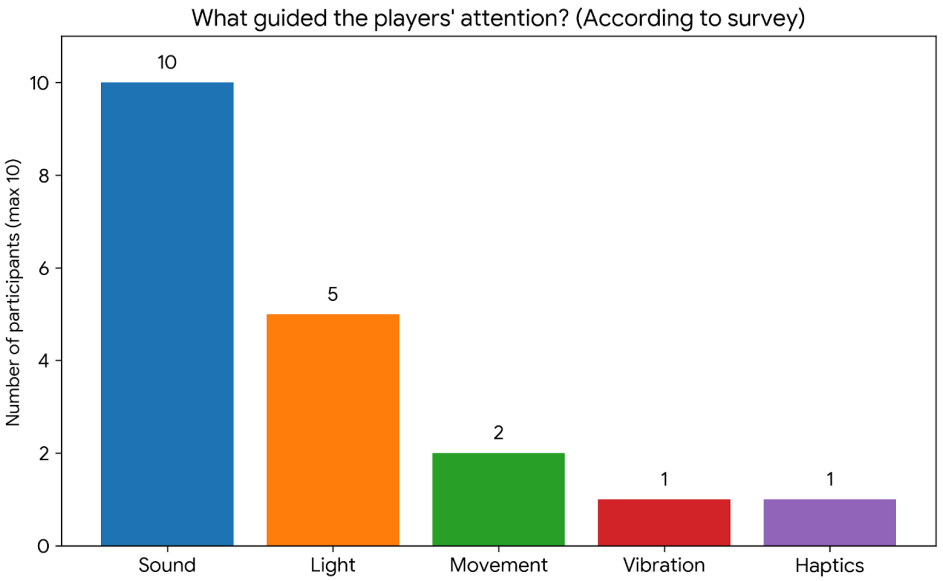


Figure 3. Graph showing the results of what guided the players attention during the experience.

The survey also asked the participants to rate how important each object felt to the overall story. They used a scale from 1 (not important) to 5 (super important). The results here were very clear. The phone, the light switch, and the painting got the highest score. Most participants gave them a 4 or a 5. The couch and the ceiling got the lowest scores, mostly receiving 1s and 2s.

In the recall task, most participants correctly identified the telephone and painting as key narrative elements. Nine out of ten participants mentioned the telephone in their descriptions, while eight mentioned the painting. In contrast, only a small number of participants (n = 2) mentioned the environmental objects such as the couch when describing the experience.

4.3 Comparing Gaze Behaviour with Narrative Importance

To see how the eye-tracking data matches the survey answers, both datasets were plotted in a scatter diagram. The image below shows the average dwell time on the X-axis and the perceived importance on the Y-axis (see figure 4).

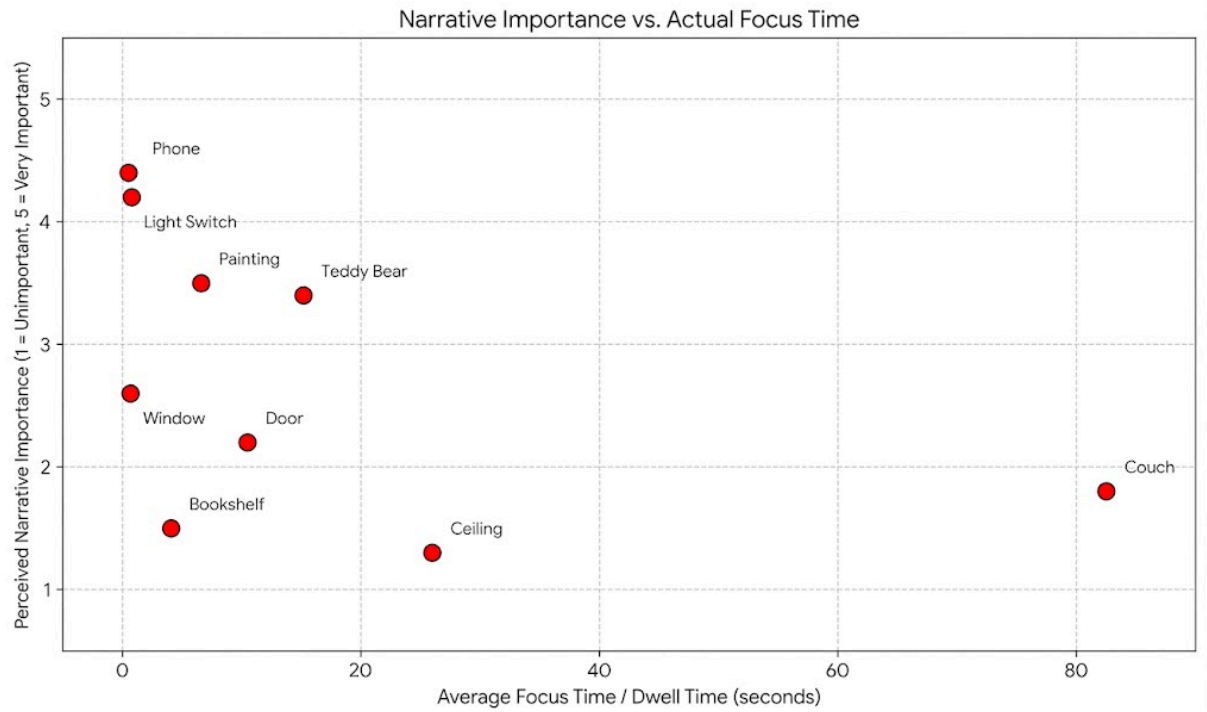


Figure 4. An image showing the narrative importance vs actual focus time of objects in the experience.

The graph above shows a clear split between physical gaze and human memory. The phone sits at the very top of the importance scale but at the very bottom of the dwell time scale. It was ranked as the most important object, but the participants barely looked at it according to the hardware.

The couch shows the exact opposite result. It sits at the far right of the dwell time scale with 82.5 seconds. But it sits at the bottom of the importance scale. Participants spent the most time looking at it, but they remembered it as almost completely unimportant to the story (See figure 5).

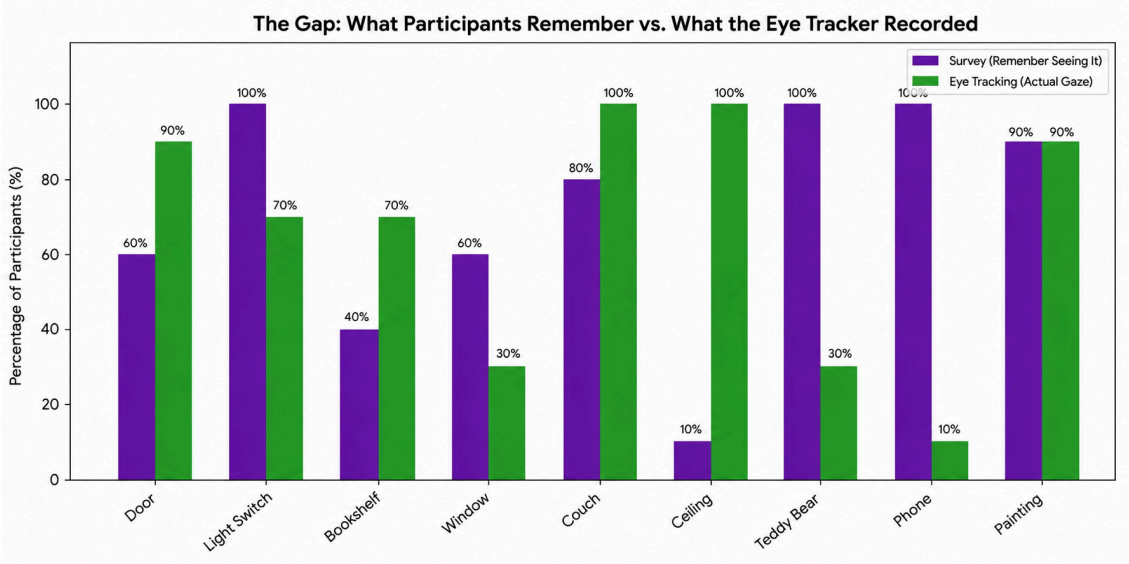


Figure 5. An image showing the difference between what the participants remember and what the eye tracker actually recorded during the experience.

To further examine the relationship between objective gaze data and subjective recall, the figure above presents the percentage of participants for whom each object was registered by the eye-tracking system alongside the percentage who reported remembering the object in the post-experience survey.

The results reveal several notable discrepancies. The phone and the teddy bear were recalled by all ten participants, yet the eye-tracking system registered gaze for only 10% and 30% of participants respectively. Meanwhile the ceiling registered gaze from all ten participants while only one participant recalled it in the survey. A similar pattern was observed for the couch, which registered gaze from all participants but was recalled by 80%, and was not considered narratively significant.

The painting represents the most aligned case, with both survey recall and eye-tracking registration at 90%. The light switch showed partial alignment, with 100% survey recall and 70% eye tracking registration, suggesting that audio cues may have directed participants attention even when the eye-tracking system did not register a direct gaze hit.

5. Discussion:

5.1 The Passive Visual Anchor

The results indicate a notable gap between what players actually look at and what they remember. The hardware logged an average of 82.5 seconds of dwell time on the couch, yet participants rated it as almost completely unimportant on the survey.

However, the couch received 82.5 seconds largely because of its large physical volume and central placement in the room. When players are standing still and listening to spatial audio or waiting for the next story event, their eyes naturally need a place to rest. Large, central objects become passive visual anchors. The player is not actively analysing the couch; they are just resting their eyes on it. Because nothing narrative happens with the couch, the brain filters it out due to a lack of saliency

(Itti & Koch, 2001). This suggests that high accumulated dwell time can be a result of physical room design and human resting gaze, rather than actual narrative importance.

5.2 Peripheral Vision and Spatial Audio

The phone shows the exact opposite result. It got the highest importance score, but the hardware only logged an average of 0.5 seconds of dwell time. Every single participant stated that sound guided their attention. When the virtual phone rings, the spatial audio tells the player exactly where it is (Pausch et al., 1996).

Because the players already know the location through the audio cue, they do not need to spend time searching the room for it. When the phone teleported behind the player, the new audio cue forced them to turn around. They found it immediately and reached out to pick it up. We can say that they likely looked at the phone during this interaction. However, because the interaction was so fast and direct, the eye tracking system may not have captured these brief gaze events reliably.

This is a direct result of the low sample rate and the small collider sizes. The logging system may not have captured brief gaze events reliably. This shows that in VR, fast and sound-guided interactions can

easily create technical blind spots in the data, making highly important objects look completely ignored.

5.3 Attention and Comprehension

During the experience, the painting started whispering, and eyes appeared that looked at the player. The eye-tracking data logged this as a success. It recorded an average dwell time of 6.6 seconds of dwell time. The players clearly found the painting, but observations during testing revealed a significant limitation. The players were confused about what to do next.

Instead of walking up to the painting and putting their ear against it, many stood still or looked around for other clues. In open-ended survey responses, participants noted that they tried to hear what the whispers said, not understanding that they came from the painting. The eye-tracking data indicates the visual cue successfully captured attention, but in reality, the player lacked an affordance. They did not understand the physical interaction required to progress.

This highlights an important limitation to VR eye tracking. Eye tracking measures visual attention, but it cannot measure comprehension. A developer can look at the data and think a puzzle is too easy because everyone looked at the clue, while the players actually felt completely stuck.

5.4 Visual and Audio Cues

In a 360-degree environment, developers cannot guarantee where the player is looking. The window event used two different cues. The main cue was visual, which was a shadow moving quickly away from the window, which only triggered if the eye tracking detected the player looking directly at it. This ensures that the player actually sees the visual event, instead of it happening behind their back.

However, if 10 seconds passed without the player looking at the window, the system used a fallback. A knocking sound was played. Observations showed that players who did not look at the window before the 10 seconds had passed immediately reacted to the knock. When they heard the knock, they walked up to the window to investigate; some even reacted with genuine fear.

This setup suggests two key aspects. First, the 10-second timer acts as a safety net that keeps the narrative pacing moving forward, preventing the player from getting permanently stuck. Second, the fear response proves that the combination of gaze-triggers and audio fallbacks successfully created a strong Plausibility Illusion (Slater & Sanchez-Vives, 2016). The brain treated the virtual threat as real, regardless of whether it was triggered by sight or sound.

5.5 Behaviour and Distraction

The data also shows how a player's previous VR experience changes how they look at things. Nine out of ten players were experienced in VR, and one was a complete beginner. However, we saw a trend in those who had little to no experience.

The experience is slightly complex since it is not a game in the traditional way, where you usually have an objective or a goal. The experience was built on the user looking around and finding out the story as they progress. Because of the users being less experienced, the immersive tech was a novelty to them. Instead of following the story, they engaged primarily in exploratory interaction with the environment. For example, some players picked up the teddy bear and just stood there shaking it to see

how gravity worked. Because they were busy playing with the environment, they completely missed it when the phone rang a second time.

This exploratory behaviour disrupted the intended narrative pacing. It shows that giving players total freedom in VR can easily ruin a story if the user is not habituated to the medium. Their attention was often directed toward environmental interaction rather than narrative progression.

5.6 Technical Limitations and Logging Improvements

The eye tracking system had several technical limitations that affected the reliability of the data. The biggest issue was the sampling rate of the logging script. As seen with the teleporting phone, players quickly turned around to grab it. But human eyes move in rapid jumps called saccades (Holmqvist et al., 2011). Because the system only pulled data a few times per second, it completely missed these rapid glances during fast interactions.

Another issue was the size of the AOIs. If the 3D collider is the exact same size as the 3D model, it requires the user to be looking straight at the object, whereas human visual perception relies heavily on peripheral awareness.

For future logging systems, two things must change. First, the 3D colliders need to be much larger than the physical objects. Second, the logging script needs to save exact timestamps for narrative events. For example, the script must log exactly when the phone teleports. By putting these event timestamps right next to the gaze logs, developers can calculate the exact time it took between the event happening and the player reacting to the event.

5.7 Methodological Reflections and Mixed-Methods

If this study only used post-experience surveys, the results would have been very misleading. The survey says participants only cared about the phone, the painting, and the light switch. It makes it look like the physical room was ignored completely.

On the other hand, if we only used eye-tracking data, the conclusions drawn would have been considerably misleading. The data says the couch and the ceiling were the main focus, and that the story objects failed to get attention.

Both methods failed when using these specific settings alone; using a higher sample rate and improving the AOIs could potentially substantially enhance the relevance of eye tracking as a data collection tool.

However, currently, the system knows where the eye points, but not why. By placing the quantitative eye-tracking numbers alongside the survey answers and the researcher’s observations, a more complete understanding emerges. The findings demonstrate that measuring narrative engagement in VR requires a mixed-methods approach.

6. Conclusion:

With this study, we sought to understand whether eye-tracking data reflects the same aspect of narrative engagement as post-experience surveys in VR, more specifically, in relation to how players follow a story. By combining subjective self-reports with objective eye-tracking data, the primary goal was to better understand how different methods capture user engagement in immersive story-driven environments.

Consequently, the results suggest that eye-tracking data, as a measure of visual attention, and post-experience surveys do not fully align when measuring narrative engagement. The eye-tracking data mainly captured where participants physically looked during the experience, while the surveys reflected what participants remembered and perceived as important to the story. However, objects with high dwell time were not always considered narratively important, and sometimes key narrative elements which were remembered as significant received very low recorded dwell time. For instance, the couch received substantially higher dwell time by acting as a passive visual anchor, while the highly significant phone received almost no dwell time because players used fast peripheral vision guided by spatial audio.

The study highlighted a crucial gap between visual attention and affordance. As seen with the whispering painting, eye tracking might log a successful visual fixation, but that does not guarantee the player understands how to physically interact with the object. The research also revealed that a player's previous VR habituation heavily alters their visual focus. Novice users were easily distracted by the physics of the environment, engaging in sandbox behaviour with objects like the teddy bear, whereas experienced users aligned their attention effectively with the intended narrative cues.

These findings indicate that eye-tracking and post-experience surveys measure different aspects of the VR experience. The survey captures memory after the experience, while eye-tracking provides insight into real-time visual attention. As a result of this, neither method alone is sufficient to fully describe narrative engagement in VR.

Ultimately, this research demonstrates that the two methods complement each other by revealing different layers of how users interpret and experience a story. However, on their own, they do not necessarily show the same aspects of narrative engagement.

These findings should, however, be interpreted as exploratory, given the limited sample size of 10 participants. While the results provide useful indications of how objective gaze measures and subjective recall may differ in narrative-driven VR experiences, further research with a larger, more diverse participant group is needed to strengthen these findings.

For future work with improved eye-tracking accuracy through higher sampling rates, the study could be conducted again to test if there is more of a connection than what this study concluded. To support this, future systems should use larger Area of Interest (AOI) colliders to capture peripheral interactions, and logging scripts should include exact timestamps to perfectly sync gaze data with triggered events. Future studies could also further explore the active use of eye tracking as a narrative tool, investigating how gaze-triggered mechanics combined with time-based audio fallbacks can secure narrative pacing, similar to the window event in this study.

Another point of investigation is additional biometric behavioural measures to better understand how attention and narrative comprehension interact in immersive environments. This could help explain physical reactions to intense narrative events, such as the fear response observed at the window.

Furthermore, testing the impact of visual clutter would be highly relevant. This study used a relatively sparse room, so investigating if audio and visual cues remain effective in a highly decorated room flooded with visual noise would provide valuable insights for level design. Furthermore, expanding the participant pool to explicitly compare complete VR beginners with experienced users, and testing more complex narrative structures, could further strengthen the understanding of how immersive storytelling functions.

7. References:

Adhanom, I. B., Koppal, S., & Höllerer, T. (2023). Eye tracking in virtual reality: A broad review of applications and challenges. *Virtual Reality, 27*(2), 1–24. https://doi.org/10.1007/s10055-022-00738-z

Burdea, G. C. (1996). *Force and touch feedback for virtual reality*. John Wiley & Sons.

Bylinskii, Z., Isola, P., Bainbridge, C., Torralba, A., & Oliva, A. (2015). Intrinsic and extrinsic effects on image memorability. *Vision research*, 116, 165-178. https://doi.org/10.1016/j.visres.2015.08.015

Cater, K., Chalmers, A., & Ward, G. (2003). Detail to attention: exploiting visual tasks for selective rendering. *In Proceedings of the Eurographics Symposium on Rendering* (pp. 270-280). https://dl.acm.org/doi/10.5555/882404.882443

Clay, V., König, P., & König, S. U. (2019). Eye tracking in virtual reality. *Journal of Eye Movement Research*, 12(1). https://doi.org/10.16910/jemr.12.1.3

Duchowski, A. T. (2017). *Eye tracking methodology: Theory and practice* (3rd ed.). Springer. https://doi.org/10.1007/978-3-319-57883-5

Holmqvist, K., Nyström, M., Andersson, R., Dewhurst, R., Jarodzka, H., & Van de Weijer, J. (2011). Eye tracking: *A comprehensive guide to methods and measures*. Oxford University Press.

Itti, L., & Koch, C. (2001). Computational modelling of visual attention. *Nature Reviews Neuroscience*, 2(3), 194-203. https://doi.org/10.1038/35058500

Jerald, J. (2015). The VR book: *Human-centered design for virtual reality*. Morgan & Claypool. https://doi.org/10.1145/2792790

Kivikangas, J. M., Chanel, G., Cowley, B., Ekman, I., Salminen, M., Järvelä, S., & Ravaja, N. (2011). A review of the use of psychophysiological methods in game research. *Journal of Gaming & Virtual Worlds*, 3(3), 181–199. https://doi.org/10.1386/jgvw.3.3.181_1

Koulieris, G. A., Aksit, K., Stengel, M., Mantiuk, R. K., Mania, K., & Richardt, C. (2019). Near-eye display and tracking technologies for virtual and augmented reality. *Computer Graphics Forum*, 38(2), 493–519. https://doi.org/10.1111/cgf.13654

Mateer, J. (2017). Directing for Virtual Reality: how the VR movie director's role needs to change. *Journal of Media Practice*, 18(1), 14-25. https://doi.org/10.1080/14682753.2017.1305838

Moreno-Arjonilla, J., López-Ruiz, A., Jiménez-Pérez, J. R., Callejas-Aguilera, J. E., & Jurado, J. M. (2024). Eye-tracking on virtual reality: A survey. Virtual Reality, 28, 38. https://doi.org/10.1007/s10055-023-00903-y

Neuert, C. E., & Lenzner, T. (2016). Incorporating eye tracking into cognitive interviewing to pretest survey questions. *International Journal of Social Research Methodology*, 19(5), 501–519. https://doi.org/10.1080/13645579.2015.1049448

Nielsen, L. T., Møller, M. B., Hartmeyer, S. D., Ljung, R. C., Nilsson, N. C., Nordahl, R., & Serafin, S. (2016). Missing the point: an exploration of how to guide users' attention during cinematic virtual reality. *In Proceedings of the 22nd ACM Conference on Virtual Reality Software and Technology* (pp. 229-232). https://doi.org/10.1145/2993369.2993405

Orquin, J. L., Ashby, N. J., & Clarke, A. D. (2016). Areas of interest as a signal detection problem in eye-tracking research. *Journal of behavioral decision making*, 29(2), 103-115. https://onlinelibrary.wiley.com/doi/abs/10.1002/bdm.1867

Pausch, R., Snoddy, J., Taylor, R., Watson, S., & Haseltine, E. (1996). Disney's Aladdin: First steps toward storytelling in virtual reality. *In Proceedings of the 23rd annual conference on Computer graphics and interactive techniques* (pp. 193-203). https://doi.org/10.1145/237170.237257

Rebenitsch, L., & Owen, C. (2016). Review on cybersickness in applications and visual displays. *Virtual Reality*, 20(2), 101–125. https://doi.org/10.1007/s10055-016-0285-9

Schacter, D. L. (1999). The seven sins of memory: Insights from psychology and cognitive neuroscience. *American psychologist*, 54(3), 182. https://doi.org/10.1037/0003-066X.54.3.182

Serafin, S., Geronazzo, M., Erkut, C., Nilsson, N. C., & Nordahl, R. (2018). Sonic interactions in virtual reality: state of the art, current challenges, and future directions. *IEEE Computer Graphics and Applications*, 38(2), 31-43. https://dl.acm.org/doi/10.1109/MCG.2018.193142628

Sheikh, A., Brown, A., Watson, Z., & Evans, M. (2016). Directing attention in 360-degree video. *In Proceedings of the IBC*. https://doi.org/10.1049/ibc.2016.0029

Sitzmann, V., Serrano, A., Pavel, A., Agrawala, M., Gutierrez, D., Masia, B., & Wetzstein, G. (2018). Saliency in VR: How do people explore virtual environments?. *IEEE transactions on visualization and computer graphics*, 24(4), 1633-1642. https://dl.acm.org/doi/10.1109/TVCG.2018.2793599

Slater, M. (2009). Place illusion and plausibility can lead to realistic behaviour in immersive virtual environments. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1535), 3549-3557. https://doi.org/10.1098/rstb.2009.0138

Slater, M., & Sanchez-Vives, M. V. (2016). Enhancing our lives with immersive virtual reality. *Frontier in robotics and AI, 3,* Article 74. https://doi.org/10.3389/frobt.2016.00074

Sundstedt, V., Chalmers, A., & Martinez, P. (2008). High fidelity gaze-based contextual interaction for virtual environments. *In Proceedings of the Eurographics Symposium on Virtual Environments*. https://dl.acm.org/doi/10.1145/1185657.1185814

8. Appendix

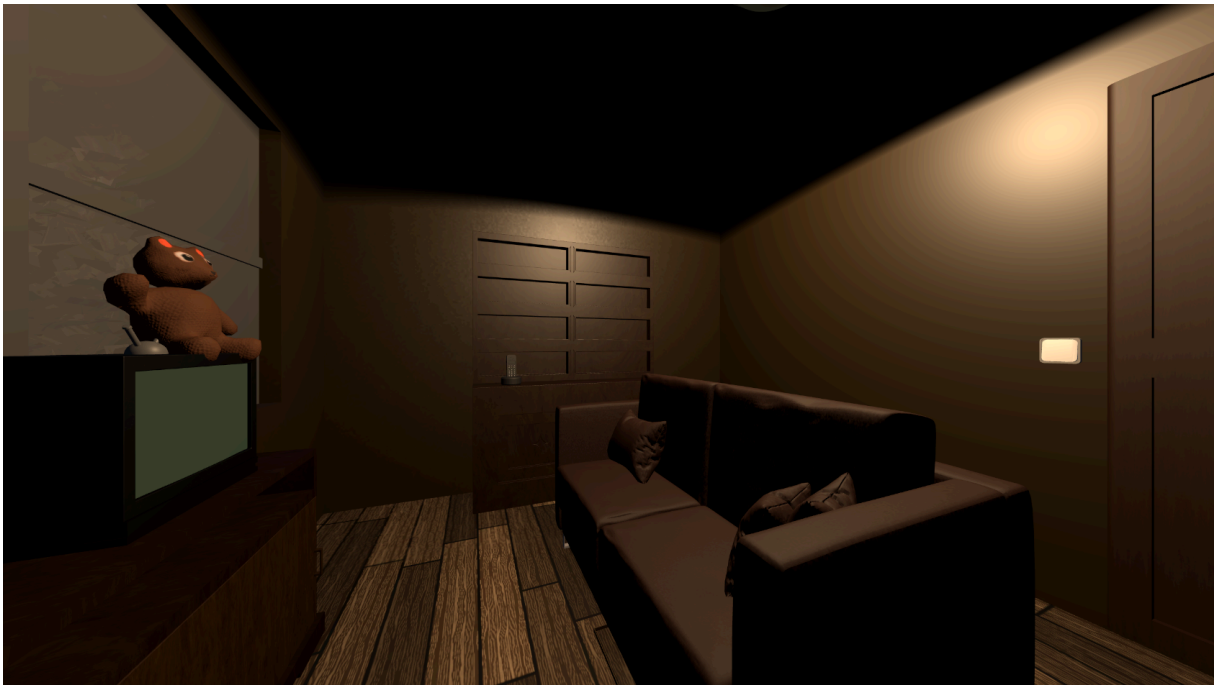


Figure 1. Overview of the narrative-driven environment showcasing the primary interactive objects.

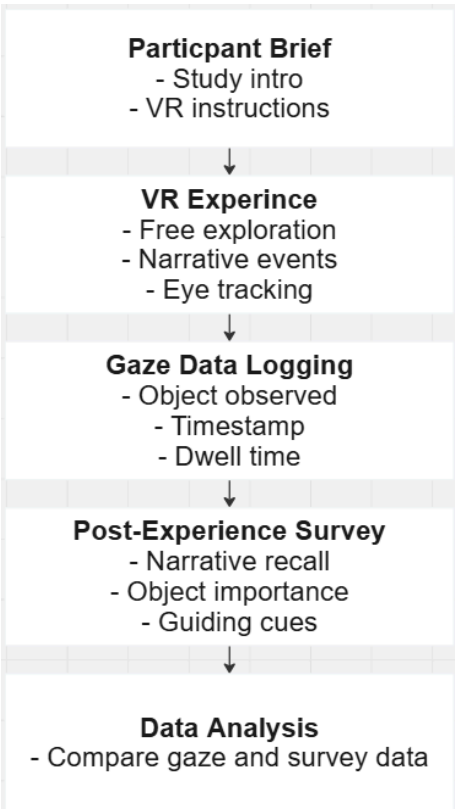


Figure 2 - An overview of the study procedure.

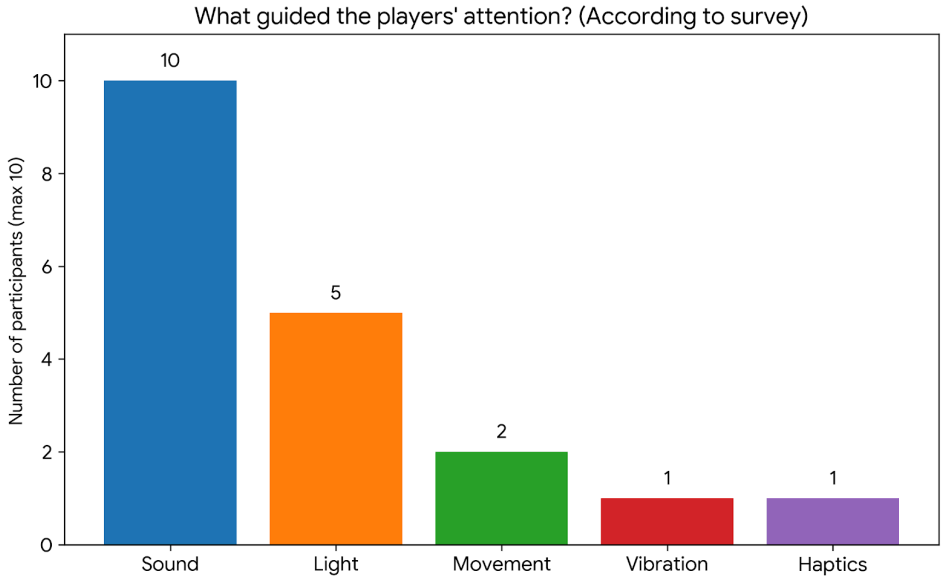


Figure 3. Graph showing the results of what guided the players attention during the experience.

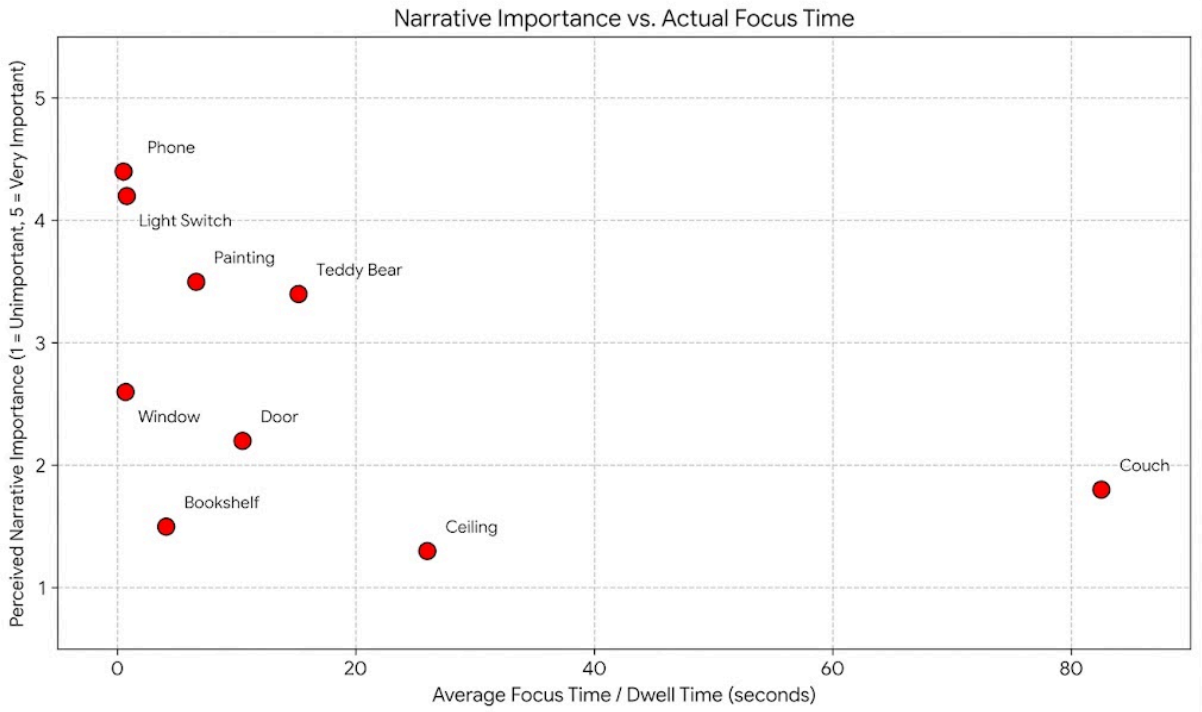


Figure 4. An image showing the narrative importance vs actual focus time of objects in the experience.

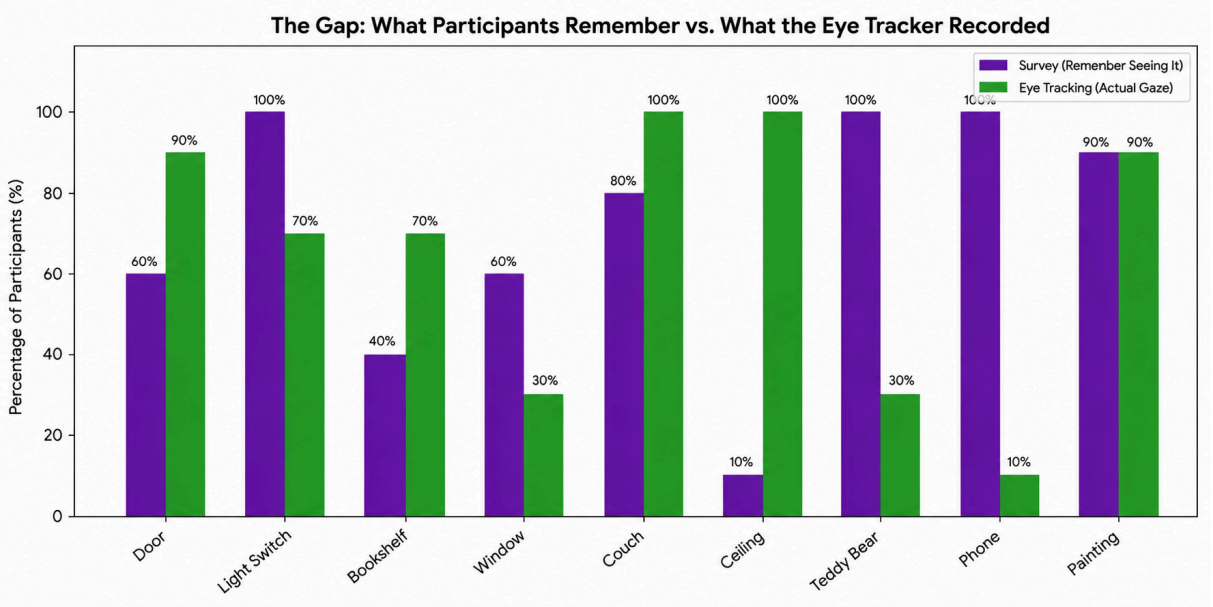


Figure 5. An image showing the difference between what the participants remember and what the eye tracker actually recorded during the experience.